UNSW MARKETING ANALYTICS HACKATHON 2026

Beyond a good story

What 1.45 million Kiva loan descriptions
can and cannot tell us about funding speed







Mike Nguyen and Sophia Nguyen
Team Cultural Blend

UNSW SYDNEY

Text carries forecast information, but no writing tip is proven

Summary across the explanatory models, robustness checks and forecast benchmark

EXPLANATORY MODELS

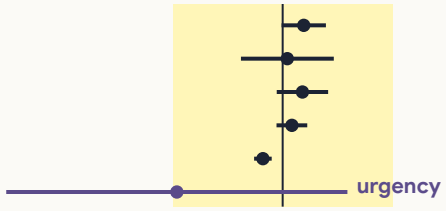
64 of 120

Uncertainty first

Yellow squares: significant terms that lost significance once errors were clustered by 48 countries.

EQUIVALENCE TEST, EXPLORATORY

±5%

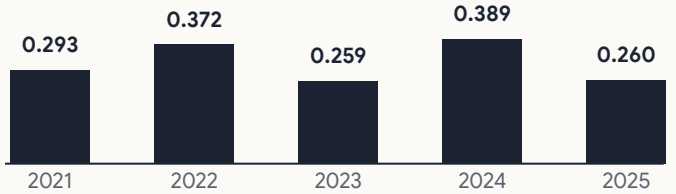


Most framing links are small

Five framings stay inside the ±5% band per extra mention per 100 words; urgency does not.

FORECAST BENCHMARK

5 of 5 years



Text adds forecast information

Text lowered linear forecast error every year, by 0.26–0.39 days; the median is about 7 hours.

Next decision: validate the text measures, then test one truthful prompt with partners.

Validate measures | Close data gaps | Pilot one prompt

Agenda

01

Problem

What are we measuring, and why does it matter?

- Editable fields 4
- The outcome 5
- A slower market 6

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Approach

How did we test the claims?

- Four analyses 7

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Evidence

What holds up, and how well?

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The description is one of the few listing fields still editable

Illustrative listing: what is fixed and what is editable when a loan is listed



FIXED AT LISTING 3

Amount, median	Term, median	Borrowers, median
US$325	**12 months**	**1**
Sector, most common	Country, most common	Partner
Agriculture	**Philippines**	**Field partner**

EDITABLE DESCRIPTION 1

She has a banana stall at the market to support her family. She wants to buy more stock quickly, before prices rise. She is thankful to every lender who helps her business grow. 4

Highlights mark framing terms counted per 100 words 2

Family | Business | Urgency | Gratitude

Figure 1. Illustrative composite listing, not a real borrower; values are dataset medians and most common categories.

1 The description is the editable part

Most fields are fixed. The **description can still be edited**, so if framing mattered, wording guidance would be a cheap lever.

2 Eight framing rates, every loan

Eight framing rates per 100 words, plus sentiment and text form, across 1,453,840 loans.

3 The test is for whom and when

Which framings are linked to **faster funding**, for which loans, and when?

4 Pronouns reflect who writes

Descriptions are mostly **written about the borrower, not by them**, so pronouns may reflect partner conventions.

The outcome is platform funding time, not borrower waiting time

Descriptive audit of all records



Loans disbursed to the borrower before listing



Who is in the data

In the data: 1,453,840 completed loans, including 1,637 refunded

Not in the data: listings that expired or never funded 2

Figure 2. Journey of a typical pre-disbursed Kiva loan, not to scale; medians from the full-data audit.

[1] Typical field-partner flow; this step is not observed in the data.

Scope: 1,453,846 input records; 6 negative durations excluded; 1,637 refunded records included.

1 Partners wait, not borrowers

96.4% of loans reached the borrower a median 24.2 days before listing; platform time plausibly reflects replenishing partner capital.

2 Unfunded loans are missing

Only loans that **eventually funded** are in the data, so we cannot study whether a loan funds at all.

3 We measure platform speed

Our outcome runs from **listing to full funding**, median 2.85 days; it is not funding success or borrower welfare.

Same-day funding fell from 46.0% to 30.3% and stayed low through 2024

Share of loans fully funded within 24 hours, by posting year; descriptive

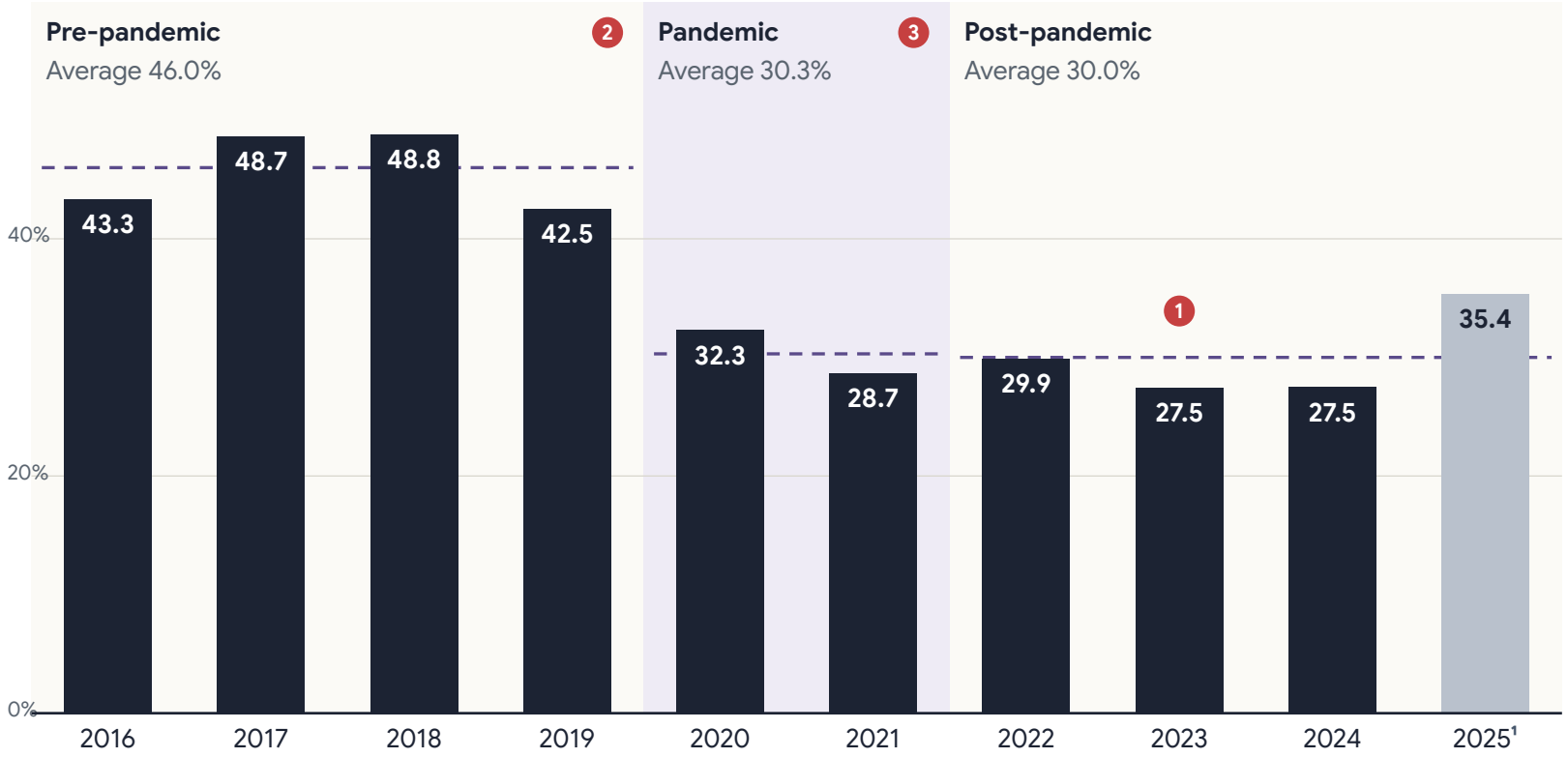


Figure 3. Share of completed loans funded within 24 hours, by posting year, with period averages.

[1] 2025 may be incomplete: loans still unfunded at extraction are missing.

1 The market stayed slower

From 2021 to 2024 the share stayed **between 27.5% and 29.9%**, well below the pre-pandemic average.

2 Every model controls for period

Period is a control in every model; **narrative effects are judged within periods**.

3 This is context, not framing

A platform-wide shift like this is **context to control for**, not evidence about how descriptions are written.

Four analyses, each answering a different question

One dataset; the equivalence test builds on the robustness checks, and the forecast benchmark asks a separate question

Kiva loans, 2016–2025: 1,453,840 valid completed durations in 48 countries

Explanatory models

Question

Which features are linked to funding speed, and how sure are we?

Method

Regression models of funding time and same-day funding, with errors clustered by country.

Data

1,453,840 loans in 48 countries

64 of 120 significant terms lost significance

Robustness checks

Question

Do the links survive richer controls?

Method

The same models with more text measures and a programme-text proxy, adjusted for multiple tests.

Data

1,453,840 loans; 642 programme categories

Only small associations remain

Equivalence test

Question

Which links are small enough to rule out a large effect?

Method

Checks whether each 90% interval sits inside ±5% of funding time.

Data

Robustness-check estimates for 9 terms

Five framings small; urgency unresolved

Forecast benchmark

Question

Does text improve a forecast built on loan structure?

Method

Train on earlier years, then test each year with and without text on the same loans.

Data

728,640 evaluation records over five years

Error fell in 5 of 5 years, by 0.26–0.39 days

Large row counts did not remove country-level uncertainty

Explanatory models: HC3 versus 48-country clustered errors

64 of 120 3

significant terms lost significance under country clustering

p = 0.49 2

urgency, duration model, clustered (was p < 0.001 under HC3)

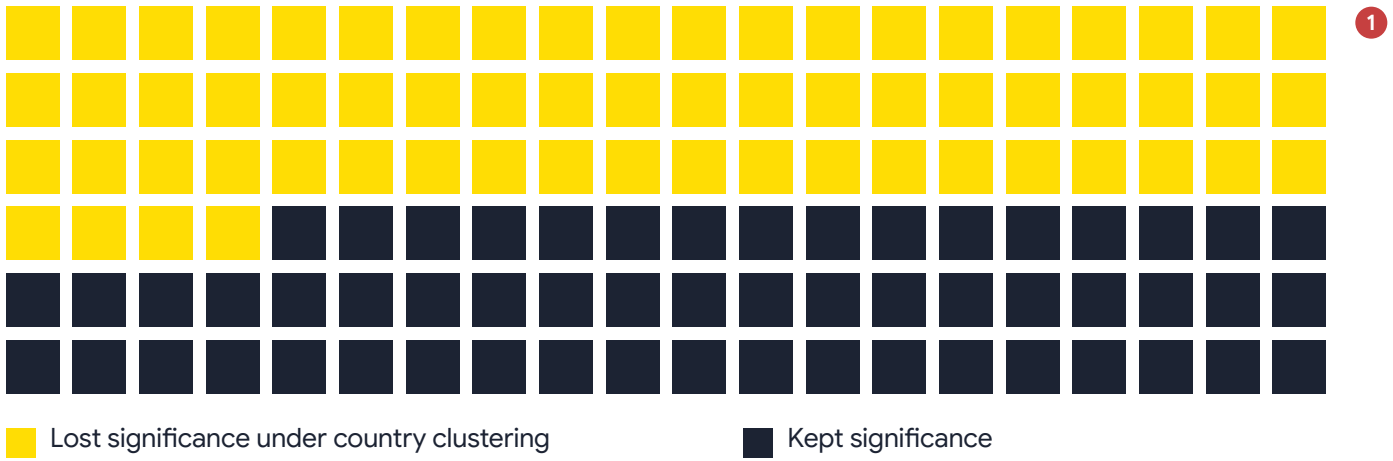


Figure 4. Terms significant under HC3 in the two explanatory models; each square is one term.

1 Countries share context

Loans in one country share context, so 1.45 million rows behave like **far fewer independent observations**.

2 Weak evidence is not no effect

Losing significance is not evidence of no effect; the next slide tests that directly.

3 48 countries act like 5.7

Ten countries hold 80% of records, so the 48 countries weigh like about **5.7 equal-sized ones** (an exploratory index).

Most framing associations are small; urgency remains unresolved

Equivalence test of the robustness-check coefficients; exploratory, bound set after the estimates

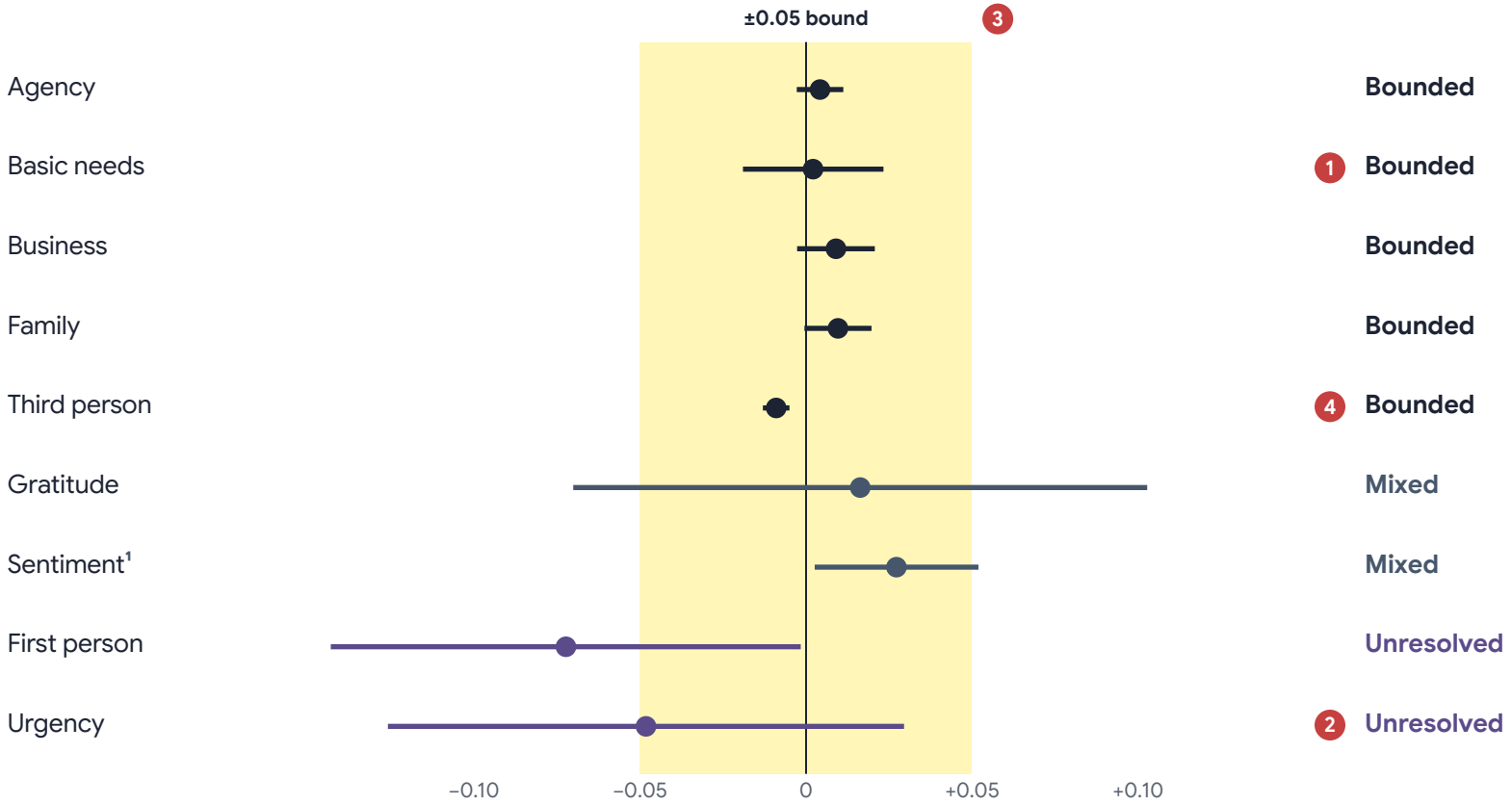


Figure 5. Change in log(1 + days) per extra mention per 100 words, 90% intervals, expanded specification; status covers both specifications.

[1] Per +0.5 change in the sentiment score.

1 Five framings are small

Five framings stay **inside ±5% of funding time** in both specifications.

2 Urgency and first person are unknown

Urgency and first-person framing could still be linked to **6–14% faster or slower funding** per mention. Each is absent from over 98% of records.

3 The bound came after the data

The bound was chosen **after seeing the estimates**; a confirmatory test needs one agreed in advance.

4 The verdicts survive a stricter test

With a stricter t(4) reference, **all five duration verdicts still hold**.

Regional family-framing patterns rest on two countries each

Explanatory models: within-region slopes with country-clustered and heuristic few-cluster intervals

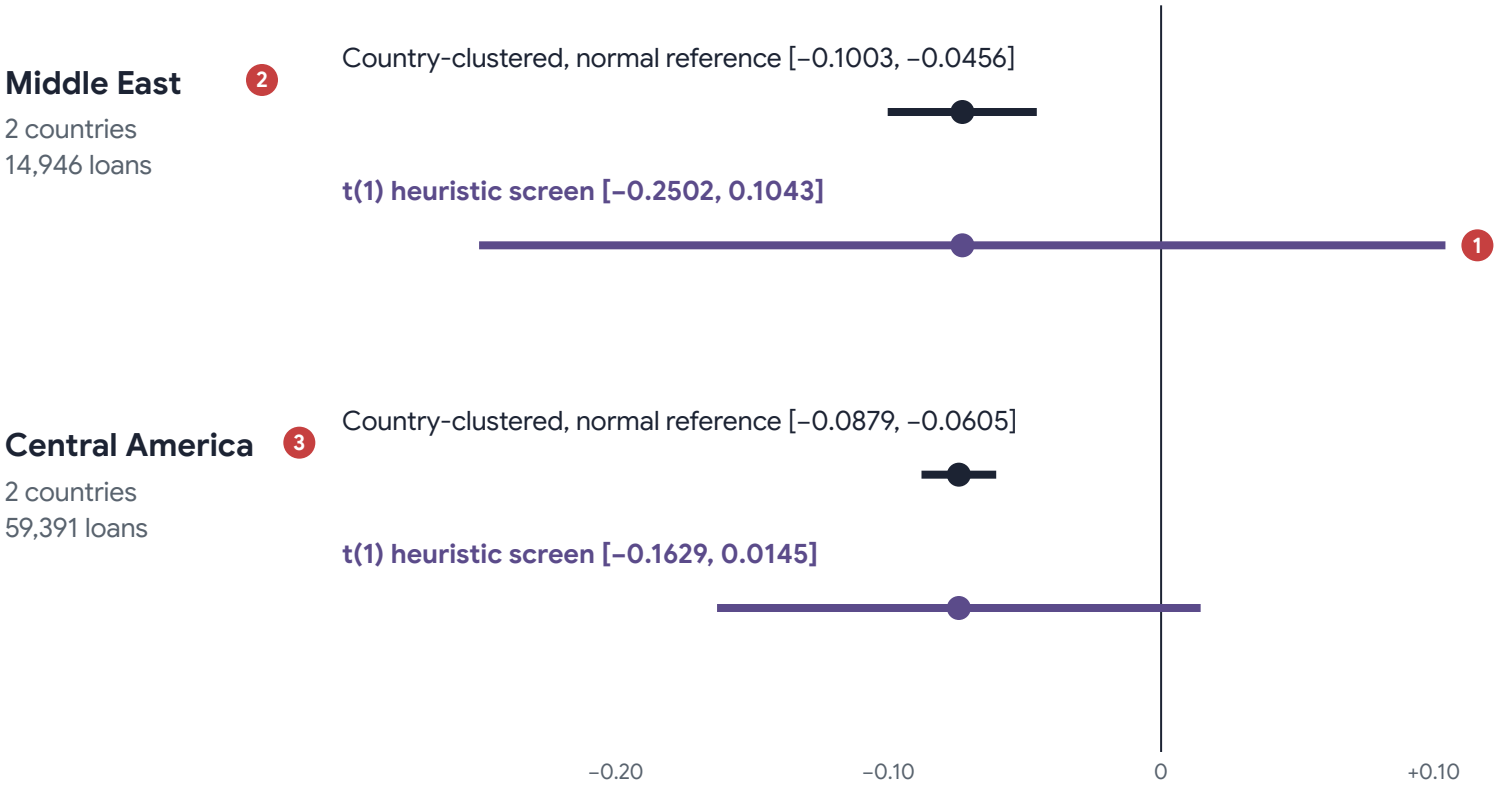


Figure 6. Within-region family-framing slopes, log(1 + days) per extra mention per 100 words.

1 Two countries cannot make a pattern

With two clusters, between-country uncertainty **cannot be estimated well**; the heuristic intervals cross zero.

2 No regional writing rules

These are descriptive patterns, **not grounds for country or regional writing rules**.

3 More countries would settle it

A regional test needs **more countries per region**, or partner-level data.

Few narrative associations remain after programme controls

Robustness checks: country-clustered, BH-adjusted within specification

Term and outcome	Expanded	Programme-adjusted
Third person, log(1 + days)	−0.0090, BH p 0.012	−0.0081, BH p 0.061
Business, 24-hour (pp)	not significant after BH	−0.258 pp, BH p 0.006

Figure 7. Per extra mention per 100 words; BH-adjusted within each specification.

0.012 → 0.061 3 1

third-person BH p, expanded to programme-adjusted

−0.258 pp 2

business framing, same-day rate, programme-adjusted

1 Only small associations remain

Third-person framing keeps its size but rises to **BH p 0.061**; business framing is **the one term below BH 0.05**.

2 These are still associations

Programme text is an **unvalidated proxy**; neither result shows what editing would do.

3 Pronouns may reflect the writer

Descriptions are usually written by partners, so the **third-person link may reflect who writes**, not a borrower's choice.

Adding text lowered forecast error in 5 of 5 years

Forecast benchmark: linear model, structure only versus structure plus text, identical evaluation rows

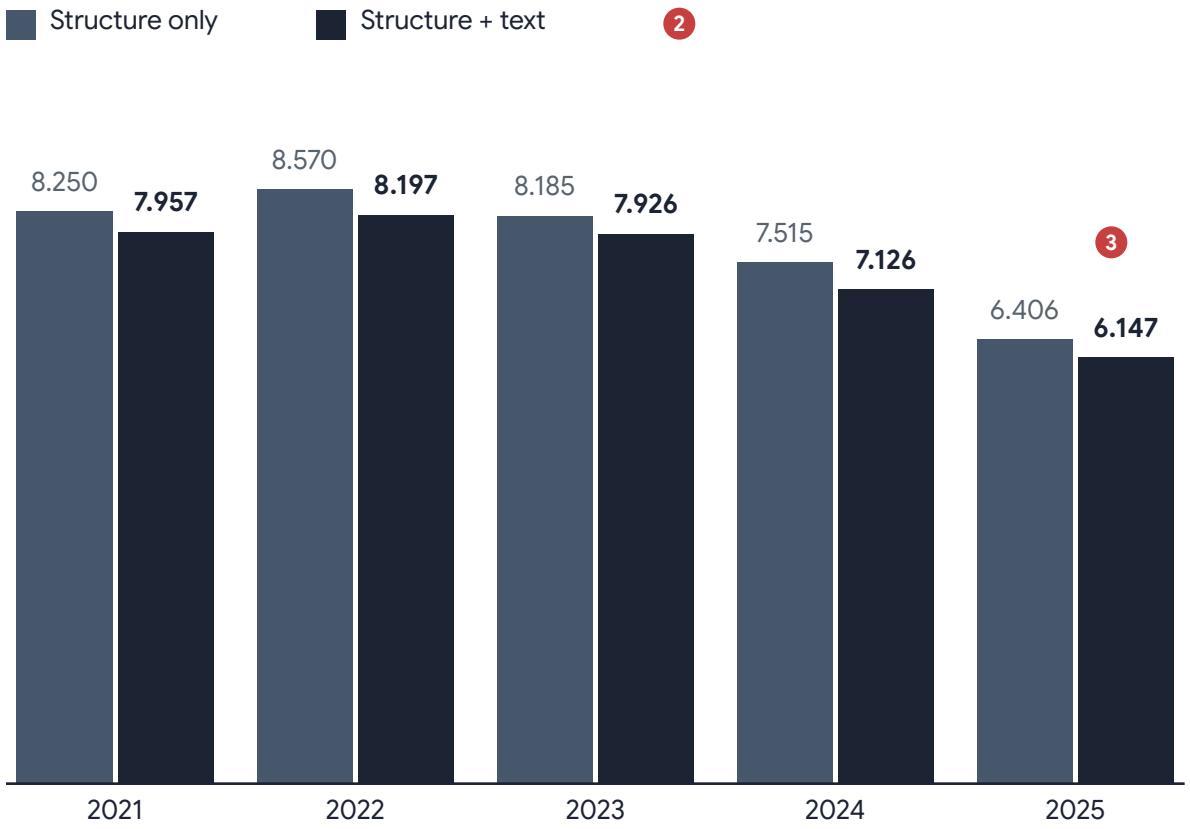


Figure 8. Linear-model MAE in days on identical evaluation rows; lower is better.

0.29 days 1

median annual error reduction

5 of 5

years with lower error

728,640

evaluation records

1 Text carries forecast information

Gains of **0.26–0.39 days** on 728,640 evaluation records.

2 Forecasting is not persuasion

Text may encode **purpose, place or programme templates**, not only persuasion.

3 No single country drives it

Removing any single country **leaves the gain positive** in every year (exploratory check).

The gain is modest and depends on the model

Forecast benchmark: annual MAE gain with month-resampled 95% intervals

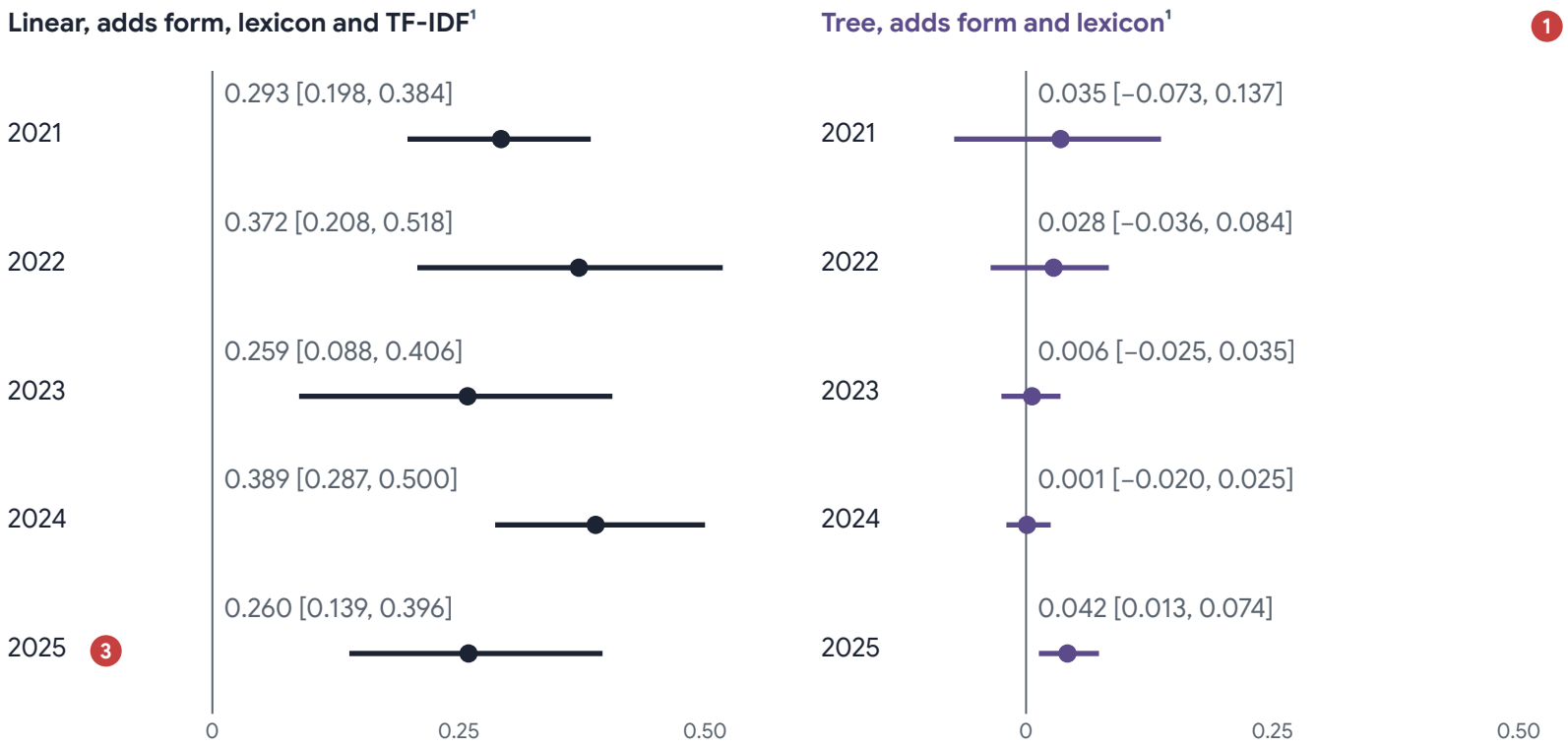


Figure 9. Annual MAE gain in days with month-resampled 95% intervals; positive is better.

2 [1] Country resampling, not shown: intervals include zero in 4 of 5 linear and 3 of 5 tree years.

1 The tree model gains little

The tree model, without TF-IDF, gains **at most 0.04 days**.

2 Country intervals are wide

Country resampling **includes zero in 4 of 5** linear years. Capping extreme forecasts, an exploratory check, leaves zero in 1 of 5.

3 Metrics move in different directions

In 2025 text improved error and calibration but **AUC fell** from 0.893 to 0.888.

What the evidence supports, and what it does not

Synthesis across all layers

Supported	Not established
Text adds forecast information in a linear model	That editing a description speeds funding
Most framing associations are small	Any country or regional writing rule
Platform funding time is not borrower waiting time for most loans	Effects of urgency or first-person framing
	Performance on loans that never funded

Three gates before any writing advice

Proposed next research for Kiva and participating partners

01

Validate

Dual-code a sample of narratives for clarity, specificity, need and agency; freeze definitions.

02

Close data gaps

Partner identifiers, all listings including expired ones, original text versions.

03

Pilot

One truthful clarity prompt versus control, with a practical threshold and power agreed in advance.

Thank you

Mike Nguyen and Sophia Nguyen, Cultural Blend

Code and evidence: github.com/tuannm3812/unsw-ma-hackathon-2026



The two-country screen in full

Explanatory models: within-region family-framing slopes in log(1 + days) per extra mention per 100 words

Region	Countries	Loans	Slope	Clustered 95%, normal	t(1) interval	Heuristic p
Middle East	2	14,946	−0.0729	[−0.1003, −0.0456]	[−0.2502, 0.1043]	0.1203
Central America	2	59,391	−0.0742	[−0.0879, −0.0605]	[−0.1629, 0.0145]	0.0597

Figure A1. Within-region family-framing slopes, log(1 + days) per extra mention per 100 words.

Why two countries are not enough

A one-country group cannot estimate between-country uncertainty, and two countries estimate it poorly. The t(1) reference keeps the same clustered standard error and widens the interval.

What the screen is

A heuristic sensitivity screen, not a calibrated correction. Passing it would not certify a result, so these patterns stay descriptive.

As-of training audit

Forecast benchmark: training records must have completed before each cutoff

Evaluation year	Pre-cutoff postings completing later	Eligible as-of training	Evaluation rows
2021	7,528	717,672	163,166
2022	3,918	884,448	143,354
2023	3,546	1,028,174	143,233
2024	5,691[1]	1,169,262	141,725
2025	4,261	1,312,417	137,162

Figure A2. Training availability and evaluation rows by year.

[1] A posting-only split would keep these 5,691 training records; evaluation windows still select eventual completions.

Chronological holdout metrics

Explanatory models: chronological holdout, completed loans only

5.2031 days

duration MAE, HistGradientBoosting

0.5420

duration R^2

0.8997

24-hour ROC AUC

0.8301

24-hour average precision

0.1156

24-hour Brier score

Scope: 1,174,953 training and 278,887 holdout loans (2024–2025). Retrospective predictions on completed loans, not deployment estimates.

Exploratory story themes

Exploratory NMF themes from a 20,000-loan sample; descriptive, not full-data evidence

1.5 days

mean funding time, fastest theme (sanitation)

13.5 days

mean funding time, slowest theme (group solar and farm plots)

What the themes show

8 themes, including sanitation, clean water, livestock, smallholder farming and retail. They track loan purpose as much as writing style.

How to read them

Means are unadjusted and come from a sample, so the gap is context, not a tested effect of story content.